Nutrition 102 (2022) 111694

Contents lists available at ScienceDirect

Nutrition

journal homepage: www.nutritionjrnl.com

Applied nutritional investigation

Comparison of generalized and athletic bioimpedance-based predictive equations for estimating fat-free mass in resistance-trained exercisers

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ARTICLE INFO

Article History:
Received 27 January 2022
Received in revised form 1 April 2022
Accepted 7 April 2022

Keywords:
BIA
BIVA
fat-free mass
predictive equations

ABSTRACT

Objectives: This study aimed to test whether athlete-specific, bioelectrical, impedance-based equations to estimate fat-free mass (FFM) could be more accurate than generalized equations when testing resistance-trained exercisers.

Methods: A total of 50 resistance-trained men (age 30.9 ± 7.4 y; body mass index: 25.3 ± 2.2 kg/m^2) and 20 men from the general population (age 29.9 ± 9.1 y; body mass index: 22.8 ± 2.4 kg/m^2) underwent bioelectrical impedance and dual-energy x-ray absorptiometry (DXA) evaluations. FFM was derived by one bioelectrical impedance-based equation specific for athletes and three generalized equations, all developed with foot-to-hand bioimpedance technologies at a 50 kHz frequency. DXA was the reference method for the FFM assessment.

Results: Compared with DXA, when assessing the resistance-trained participants, the athletic-specific equation had neither mean (−0.89 kg; $P = 0.789$) or proportional bias ($r = -0.104$; $P = 0.474$) with a coefficient of determination equal to $R^2 = 0.91$. In contrast, the three generalized predictive equations overestimated FFM (range, 4.11−5.37 kg; $P < 0.05$) with R^2 ranging from 0.84 to 0.90. The athletic-specific equation underestimated FFM in the general population participants (−2.93 kg; $P < 0.05$).

Conclusions: When assessing body composition in resistance-trained exercisers, specific equations for athletes should be preferred to generalized ones to avoid an overestimation in FFM. Furthermore, athlete-specific and generalized formulas cannot be used interchangeably, even when assessing body composition in the general population.

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Introduction

Current classifications identify athletes as subjects engaged in physical activity with the primary objective of improving performance to support athletic excellence and/or success [1−3]. In addition, athletes are only considered to be those who compete in organized competitions, with at least a minimum of 2.5 h of training per wk. Otherwise, all other sports practitioners are identified as exercisers (i.e., subjects who participate in physical activity with the motivation to increase fitness, promote health, improve physique, and learn or refine skills) [1−3].

Exercisers, such as resistance-trained subjects, are frequently included as participants in studies aiming to assess the effectiveness of nutritional interventions or training strategies, and body composition is a commonly evaluated parameter [4,5]. Among the methods developed to assess body composition, the bioelectrical impedance analysis (BIA) represents an easy and feasible technique to obtain a qualitative and quantitative assessment [6], and is used frequently

Conflicts of Interest: FJT and CNM are currently employees of a biotechnology company (Bettery, S.A.) that produces dietary supplements; however, no funding or resources of this company were used in this investigation.

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https://doi.org/10.1016/j.nut.2022.111694
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in the sports context in contrast with more accurate but less available methods [7]. In this regard, BIA and bioelectrical impedance spectroscopy (BIS) at 50 kHz have been proposed as valid tools compared with more expensive techniques, such as underwater weighing air plethysmography, dual-energy x-ray absorptiometry (DXA), or dilution techniques [6,8–11]. To date, BIS and BIA represent two double-indirect methods based on the assessment of body impedance [6,12,13]. The BIS method was the first proposed approach to measure impedance, and uses a range of frequencies from 4 to 1000 kHz. The derived parameters are used in nonlinear mathematical models to estimate bioelectrical components [14]. In contrast, when using phase-sensitive devices, the impedance can be separated into bioelectrical resistance (R) and reactance (Xc) at different frequencies [15]. Specifically, the qualitative analysis allows for the evaluation of R and Xc by plotting a vector within a graph and comparing the vector with the reference values of different populations [9]. On the other hand, a quantitative analysis, based on the relationship between bioelectrical values and body composition features, allows for the estimation of parameters, such as fat-free mass (FFM), fat mass, or body fluids, through prediction equations [9].

The current literature proposes a wide range of prediction equations to quantify body composition components, available for both athletic and nonathletic populations [8,16]. FFM is one of the most monitored parameters given its relationship with sports performance [17,18], and specific equations for both athletes [19,20] and the general population are currently available [21–23]. However, apart from the intent of sports practice, which is considered the main discriminant for the classification of a subject as an athlete [3], exercisers can show body composition features similar to those of athletes. For example, some resistance-trained exercisers involved in previous research showed similar body composition features to those of athletes and far from the average values of the general population [5,24,25].

In this regard, the continuous resistance training practice (at least 3 h/wk) can lead to the development and maintenance of FFM [5,24]. Resistance training is a form of exercise designed to increase muscle mass and strength by exercising against external resistance, such as barbells, cables, or weight machines [5]. Recently, some authors proposed that, when assessing body composition in resistance-trained exercisers, equations for athletes could be more accurate than generalized formulas [2]. However, no previous study has tested this hypothesis. Specific bioimpedance-based equations [19,20] have been shown to provide FFM values similar to those obtained with reference methods when applied in athletes [10]. Furthermore, although the underestimation of FFM in athletes when using general equations is known [8], the consequence of using athletic-equations in subjects from the general population is still unexplored.

Therefore, the aim of the present study was to test whether or not bioimpedance-based equations for athletes compared with the generalized population show better agreement in the assessment of FFM, with DXA as the reference method, when resistance-trained men are tested. In addition, a group of subjects from the general population was used as the control group. Our hypotheses were that athletic-specific equations yield better predictions of body composition compared with general equations in exercisers, and generalized equations show a better agreement with DXA than athletic-specific formulas when assessing the general population.

Methods

Study design and participants

The study was conceived as a cross-sectional, two-group investigation. Body composition assessments were performed on the same day around 09:00 h using BIS and DXA. Bioelectrical impedance-derived FFM was estimated using a specific equation developed for athletes involved in different sport modalities [20] and three generalized formulas [21–23] as follow:

Equation for athletes [20]:

$$\text{FFM}_{\text{Matias}} = -2.261 + 0.327 \times \text{height}^2/\text{R} + 0.525 \times \text{body mass} + 5.462 \times \text{sex, where sex is 1 for men and 0 for women}$$

Generalized equation 1[22]:

$$\text{FFM}_{\text{Lohman}} = (0.73 \times [\text{height}^2/\text{R})] + (0.16 \times \text{body mass}) + 2.0$$

Generalized equation 2 [23]:

$$\text{FFM}_{\text{Lukaski}} = 0.734 \times \text{height}^2/\text{R} + 0.116 \times \text{body mass} + 0.096 \times \text{Xc} + 0.876 \times \text{sex, where sex is 1 for men and 0 for women}$$

Generalized equation 3 [21]:

$$\text{FFM}_{\text{Sun}} = -10.68 + 0.65 \times \text{height}^2/\text{R} + 0.26 \times \text{body mass} + 0.02 \times \text{R}$$

All selected equations were developed with bioimpedance foot-to-hand technologies at a frequency of 50 kHz. The athletic-specific equation was selected as the only one available in the literature to quantify FFM in athletes. Otherwise, the three generalized equations were selected because they have already been used in previous sports-related studies [8].

A total of 50 resistance-trained men (age: 30.9 ± 7.4 y; body mass index: 25.3 ± 2.2 kg/m^2) and 20 active men from the general population (age: 29.9 ± 9.1 y; body mass index: 22.8 ± 2.4 kg/m^2) voluntarily participated in this observational study. Resistance-trained participants were experienced in regular resistance training for at least 2 y, and practiced whole-body hypertrophy-type resistance training routines for three training sessions per week, with a minimum of 48-h intervals between sessions. The resistance-trained men were tested for one repetition maximum (1RM) for back squat and bench press exercises: 1RM squat = 135.4 ± 30.4 kg; 1RM bench press = 90.4 ± 30.1 kg.

The exclusion criteria were comprised of muscle injury during the previous 6 mo or participating in organized competitions. General population participants were selected using the following inclusion criteria: Nonathletes, not practicing resistance training for >2 y or three training sessions per week, and no injuries or illnesses at the time of the test. After a detailed explanation of the procedures, participants signed an informed consent. All research procedures were reviewed and approved by the Faculty of Human Kinetics institutional review board, and conformed to the Declaration of Helsinki (1964 and further updates) concerning studies involving human subjects.

Body composition assessment

All athletes were tested to ensure a well-hydrated state using the urine-specific gravity test (refractometer Urisys 1100; Roche Diagnostics, Lisbon, Portugal) from a fasting baseline urine sample according to the study by Armstrong et al. [26]. A urine-specific gravity value of <1.022 in the first urine sample was used to categorize euhydration. Height was recorded to the nearest 0.1 cm with a stadiometer, and body mass was measured to the nearest 0.1 kg with a high-precision scale (Seca, Hamburg, Germany). DXA was performed on a Hologic Explorer-W, fan-beam densitometer (Hologic, Waltham, MA) to obtain total whole-body FFM. The DXA scan was calibrated with a soft-tissue phantom before the test session.

The bioimpedance parameters (R and Xc) were directly measured with a multi-frequency analyzer (BIS 4200, Xitron Technologies, San Diego, CA) at a frequency of 50 kHz. The bioelectrical phase angle was calculated as the arctangent of $\text{Xc}/\text{R} \times 180°/\pi$. The accuracy of the bioimpedance analyzer was validated with a resistor before the test session, and measurements were performed according to standardized procedures [6]. Whole-body R and Xc were also analyzed according to the bioelectrical impedance vector analysis (BIVA) to qualitatively assess body composition [27]. The R–Xc graph was used to plot the BIVA data by comparing them to the athletic [28] and general [29] reference populations.

Statistical analysis

Data were analyzed with SPSS, version 27.0 (SPSS, IBM Corp., Armonk, NY) and MedCalc statistical software, version 11.1.1.0, 2009 (Mariakerke, Belgium). The Shapiro–Wilk test was used to check the normal distribution of the data. A paired sample *t* test was employed to compare the FFM obtained from DXA and the BIA for each equation. A linear regression analysis was performed considering the values obtained from DXA as dependent variables and the bioelectrical impedance-derived parameters as independent variables. Agreement between the bioelectrical impedance-based equations and DXA was determined using the Bland–Altman method and Lin's concordance correlation coefficient (CCC), including precision (ρ) and accuracy (C_b) indices, and classified according to McBride's strength

concordance (almost perfect: CCC >0.99; substantial: CCC <0.99 to >0.95; moderate: CCC <0.95 to >0.90; and poor: CCC <0.90).

The independent sample Hotelling's T^2 test (BIVA software [30]) was used to compare the mean impedance vectors of the participants with the reference data for the athletic and general populations [28,29], where the statistical significance was corrected for the number of analyses and defined at $P < 0.0125$. Mahalanobis distance (D^2), which represents a multivariate measure of effect and multivariate measure of distance, was calculated to determine the magnitude of the difference in the mean vectors. Threshold values were identified for $D^2 < 0.5$ as small, for $0.5 \leq D^2 < 0.8$ as medium, and for $D^2 \geq 0.8$ as large effect.

Results

BIVA patterns are depicted in Figure 1, where a comparison with the reference 50%, 75%, and 95% tolerance ellipses of the athletic and general population is also shown. The results of the two-sample Hotelling T^2 test showed significant differences for the mean vectors between the two group of participants and the reference values for the athletic and general populations (Fig. 1). The resistance-trained participants' mean vectors with their 95% confidence ellipse fell partly outside of the 95th percentile of the reference values for the general population. On the contrary, the mean vector of the resistance-trained men was included between the 50th and 95th percentiles of the athletic population. The mean vector of the general population participants was included among the 50th percentile of the general population and between the 75th and 95th percentiles of athletes' reference tolerance ellipses. The mean vectors of the general participants did not differ from the reference values of their population, as reported in Figure 1.

Table 2 and Supplementary Figures 1 and 2 present the results from the regression, CCC, and agreement analysis between the FFM estimated from the selected predictive equations and the reference method.

Concerning the resistance-trained men, no difference was observed between FFM estimated from the Matias equation [20] and DXA ($P = 0.789$). All other predictive equations overestimated FFM compared with DXA ($P < 0.05$). Also, the regression analysis showed that FFM estimated from the Matias equation [20] was highly correlated with the reference method estimation (91% power explanation), but the other predictive equations presented a power explanation ≤90% (Table 1; Suppl. Fig 1). Likewise, the standard error of estimation corroborates the aforementioned data, because the FFM predicted from the Matias equation [20] presented the lower error in the estimate.

Regarding the CCC analysis, a substantial agreement between methods [31] was observed between the Matias equation [20] and DXA to predict FFM, but the other equation presented a poor strength of agreement (Table 1; Suppl. Fig 1). At an individual level, according to the Bland–Altman agreement analysis, no trend was observed in any of the predictive equations; nevertheless, limits of agreement were lower (~7% of total mean FFM) when using the Matias equation [20] (Table 1; Suppl. Fig 1). The FFM of participants from the general population was underestimated with the Matias equation [20], but no difference was observed between the DXA and bioimpedance outcomes when the generalized equations were applied (Table 2; Suppl. Fig 2). Limits of agreement for the aforementioned three equations were from 8.2% to 10.8% of the mean FFM.

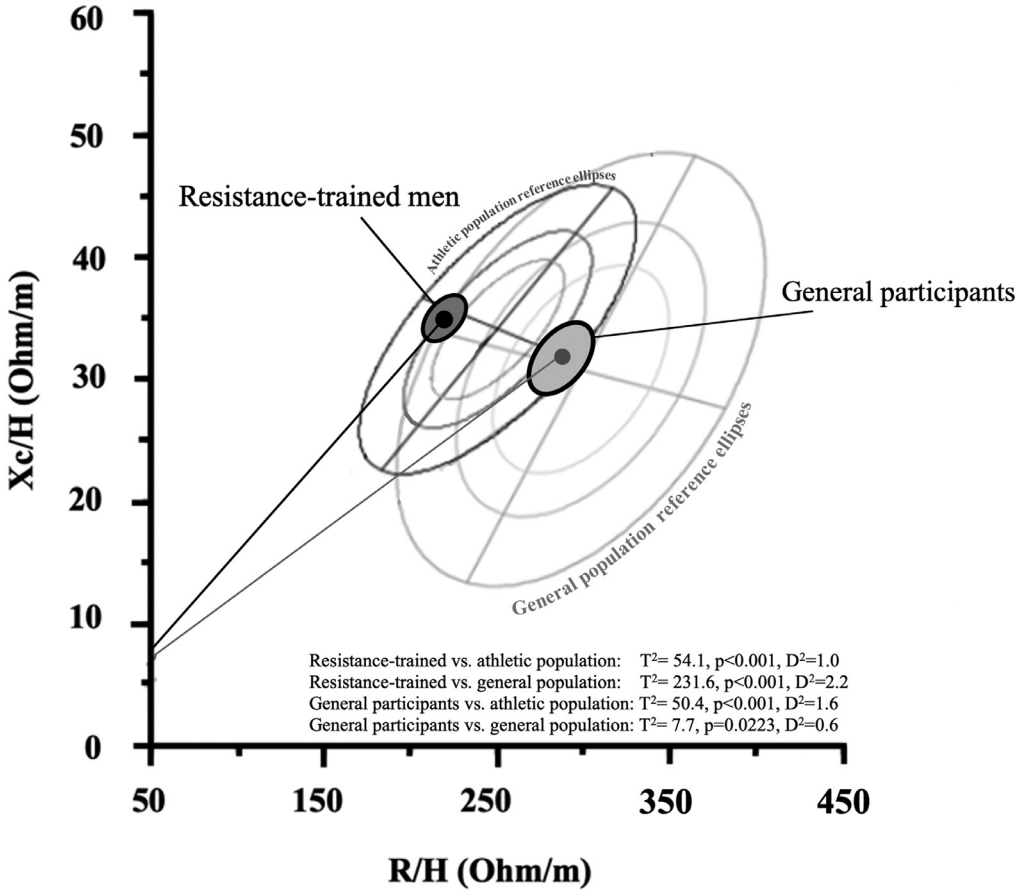

Fig. 1. Mean vector with 95% confidence ellipse of resistance-trained and general population participants in comparison with tolerance reference ellipses of athletic [28] and general populations [29]. Results of Hotelling's test are also shown.

Table 1
Descriptive characteristics of participants

Variable	Resistance-trained men Mean ± SD	General population men Mean ± SD
Height, cm	174.0 ± 5.8	181.6 ± 5.1
Body mass, kg	76.3 ± 9.5	75.1 ± 9.0
Resistance, ohm	396.3 ± 45.7	517.1 ± 50.0
Reactance, ohm	59.0 ± 7.1	60.2 ± 6.5
Phase angle, °	8.5 ± 0.7	6.6 ± 0.3
FFM_{DXA}, kg	63.7 ± 7.4	61.2 ± 6.2
FFM_{Matias} [15]	63.6 ± 7.7	58.2 ± 6.4
FFM_{Lohman} [17]	69.1 ± 7.6	61.9 ± 6.0
$FFM_{Lukaski}$ [18]	68.5 ± 8.2	61.0 ± 6.0
FFM_{Sun} [16]	67.8 ± 8.1	61.1 ± 5.9

DXA, dual-energy x-ray absorptiometry; FFM, fat-free mass; SD, standard deviation. Data are reported as mean ± SD.

Discussion

The present study aimed to test whether athlete-specific, bioelectrical, impedance-based equations to estimate FFM can be more accurate than generalized ones when testing resistance-trained exercisers. Another aim was to test the performance of athletic-specific and generalized equations compared with DXA to assess FFM in a group of general population men. The results showed a good agreement between bioimpedance and DXA only when the athletic-specific equation was used in resistance-trained participants. On the contrary, all generalized equations overestimated FFM compared with DXA. When assessing general population participants, the specific equation for athletes underestimated FFM, but the generalized formula outcomes agreed with the DXA estimated value, which suggests that resistance-trained men should be considered athletes when using a bioimpedance analysis to quantitatively assess body composition. Additionally, athlete-specific and generalized equations are not interchangeable when testing the general population.

The R−Xc graph allows for the evaluation of body fluid content and its distribution between intra- and extracellular compartments [9]. The major axis of the ellipses is an index of fluid content, where higher values are positioned at the bottom and lower values at the top. Otherwise, the minor axis is an index of the ratio between intra- and extracellular fluids and FFM, where values higher than the average are placed in the left part of the graph [32,33].

The current participants were analyzed with a BIVA, comparing their bioelectrical properties against the references of the athletic and general populations. Although significant differences between the resistance-trained participants showed a mean vector difference for both the athletic and general populations, the magnitude of the differences was higher with respect to the general population. In contrast, general population participants only differed from the athletic population.

Resistance-trained participants fell partly outside of the 95th percentile of the general population, positioning in the left quadrant of the R−Xc graph. In contrast, when considering the references of the athletic population, resistance-trained participants fell between the 50th and 95th percentiles, showing body composition features closer to those of athletes than those of the general population. Interestingly, showing a vector direction more to the left than the center of the ellipses of the athletic population, resistance-trained men may have a higher phase angle than several types of athletes, as confirmed by the phase angle reference percentiles for athletes [34]. In this regard, the current resistance-trained men not only showed higher values in the intra-to-extracellular water ratio and FFM compared with the general population, but even higher on average than the athletic population.

In line with our results, previous studies have shown that groups of exercisers may show BIVA patterns and therefore body composition characteristics similar to those of the athletic population [5,35]. In particular, previous studies applied BIVAs on amateur cyclists and endurance-trained men, showing a vector distribution within the 50th percentile of the R−Xc graph when the reference tolerance ellipses of the athletic population were used [5,35]. When compared with the BIVA references, participants belonging to the general population positioned within the 50% tolerance ellipse of their reference population and far from the average percentiles of athletes, which is in line with previous findings where athletes and subjects belonging to the general population showed different BIVA patterns due to a greater phase angle in subjects with a higher degree of training [28,36].

Our results show that the Matias equation [20] was the most adequate method to predict FFM in resistance-trained men. The better agreement of the athletic-specific equation may be due to the fact that resistance-trained exercisers showed, on average, FFM values (63.2 kg) similar to those of athletes involved in the Matias study [20] (FFM = 65.2 kg), where the athletic-specific equation was developed. Bearing in mind that this equation was developed for an athletic population, this result corroborates the aforementioned idea that groups of exercisers may show body composition characteristics similar to those of athletic populations,

Table 2
Validation of regression equations in resistance-trained and general population participants

		Regression analysis		CCC analysis			Agreement analysis		
	Mean ± standard deviation	Coefficient of correlation	Standard error of estimation, kg	CCC	Precision	Accuracy	Bias	95% limits of agreement	Trend
Resistance-trained men (n = 50)									
FFM_{DXA}	63.7 ± 7.4	–	–	–	–	–	–	–	–
FFM_{Matias} [15]	63.6 ± 7.7	0.91	2.29	0.951	0.952	0.999	−0.89	−4.68; 4.50	r = −0.104; P = 0.474
FFM_{Lohman} [17]	69.1 ± 7.6	0.90	2.40	0.751	0.947	0.736	5.37*	0.55; 10.19	r = 0.097; P = 0.501
$FFM_{Lukaski}$ [18]	68.5 ± 8.2	0.84	3.02	0.762	0.915	0.833	4.82*	−1.66; 11.31	r = 0.247; P = 0.084
FFM_{Sun} [16]	67.8 ± 8.1	0.89	2.52	0.871	0.941	0.820	4.11*	−1.22; 9.44	r = −0.241; P = 0.091
General population participants (n = 20)									
FFM_{DXA}	61.2 ± 6.2	–	–	–	–	–	–	–	–
FFM_{Matias} [15]	58.2 ± 6.4	0.82	2.71	0.812	0.905	0.897	−2.93*	−8,32; 2,46	r = 0.069; P = 0.777
FFM_{Lohman} [17]	61.9 ± 6.0	0.78	3.03	0.871	0.880	0.990	0.79	−5,07; 6,66	r = −0.086; P = 0.718
$FFM_{Lukaski}$ [18]	61.0 ± 6.0	0.77	3.06	0.876	0.877	0.990	−0.15	-6,08; 5,78	r = −0.075; P = 0.754
FFM_{Sun} [16]	61.1 ± 5.9	0.75	3.17	0.866	0.867	0.998	0.09	−6,22; 6,05	r = −0.102; P = 0.669

CCC, concordance correlation coefficient; DXA, dual-energy x-ray absorptiometry; FFM, fat-free mass.
*Significant differences with reference method ($P < 0.05$).

and therefore are more closely related to the methods developed for these than those for general population predictive equations or methods.

Every day, bioimpedance technology and its predictive models are more specific for several populations. Therefore, the use of a nongeneralized formula could be accurate to measure groups of exercisers, in whom high values of FFM may be due to the practice of resistance training even if contained in <3 h per wk [37]. In this regard, generalized equations, such as the Lukaski equation [23], have been developed on a general population with very different characteristics from the athletic population (FFM = 72.6 kg). However, the Lukaski equation [18] proved to be valid for the FFM assessment even when applied to groups of athletes, when its performance was tested against the hydrodensotometry technique [38]. Otherwise, the majority of the generalized equations lead to a lack of agreement between reference methods and bioimpedance when quantifying body fluids in athletes [8]. Therefore, although generalized equations should be avoided for athletes, equations for athletes possibly are more appropriate in groups of exercisers to quantify body composition.

The main strength of the present study is the sample group of resistance-trained men not participating in any organized competition, considering them as exercisers rather than athletes. Additionally, this is the first study to show how equations for athletes and generalized formulas cannot be used interchangeably, even in the general population. The resistance-trained men involved in this study had a consistent background of resistance training, and their body composition characteristics were more similar to those of athletes rather than to those of the general population. This allowed for testing of the hypothesis that in some cases, athletic-specific equations should be applied, even when individuals cannot be defined as such.

Some study limitations should also be mentioned. Our results cannot be applicable to female resistance-trained exercisers. In addition, only FFM was considered, and our findings cannot be extended to other body composition variables, such as body fluids or muscle mass. Therefore, further studies should include female exercisers, as well as consider other body composition component. Furthermore, these findings are related only to the specific equations assessed, and the weakness of the generalized equations cannot be extended to all published equations or all populations. Finally, the bioelectrical outcomes were derived from the use of a foot-to-hand technology and a 50-kHz sampling frequency. Therefore, the present results may not be used interchangeably with different technologies (e.g., multifrequency BIA in standing position) and sampling frequencies, including devices using unknown equations owned by the manufacturers.

Conclusions

From an applied point of view, researchers and practitioners should first analyze BIVA patterns in the tested exercisers, and when these are similar to those of the athletic population, specific formulas for athletes should be preferred over the generalized ones. Future research should apply a similar experimental design by considering another low-cost method, such as anthropometry, suggesting how anthropometrists should act when resistance-trained exercisers are measured and anthropometry-based equations are applied. Additionally, raw bioelectrical values, such as phase angle, should be studied with the purpose of defining specific cut-off points aimed at guiding practitioners toward using an equation for athletes rather than a generalized one.

Groups of exercisers, such as resistance-trained men, may exhibit similar body composition characteristics to those of athletes. Therefore athlete-specific equations should be preferred over general ones when quantifying body composition variables using a BIA at 50 kHz. In this regard, using general equations in resistance-trained exercisers leads to an overestimation in predicted FFM.

Acknowledgments

The authors express their gratitude to all subjects who volunteered for this study.

Supplementary materials

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.nut.2022.111694.

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